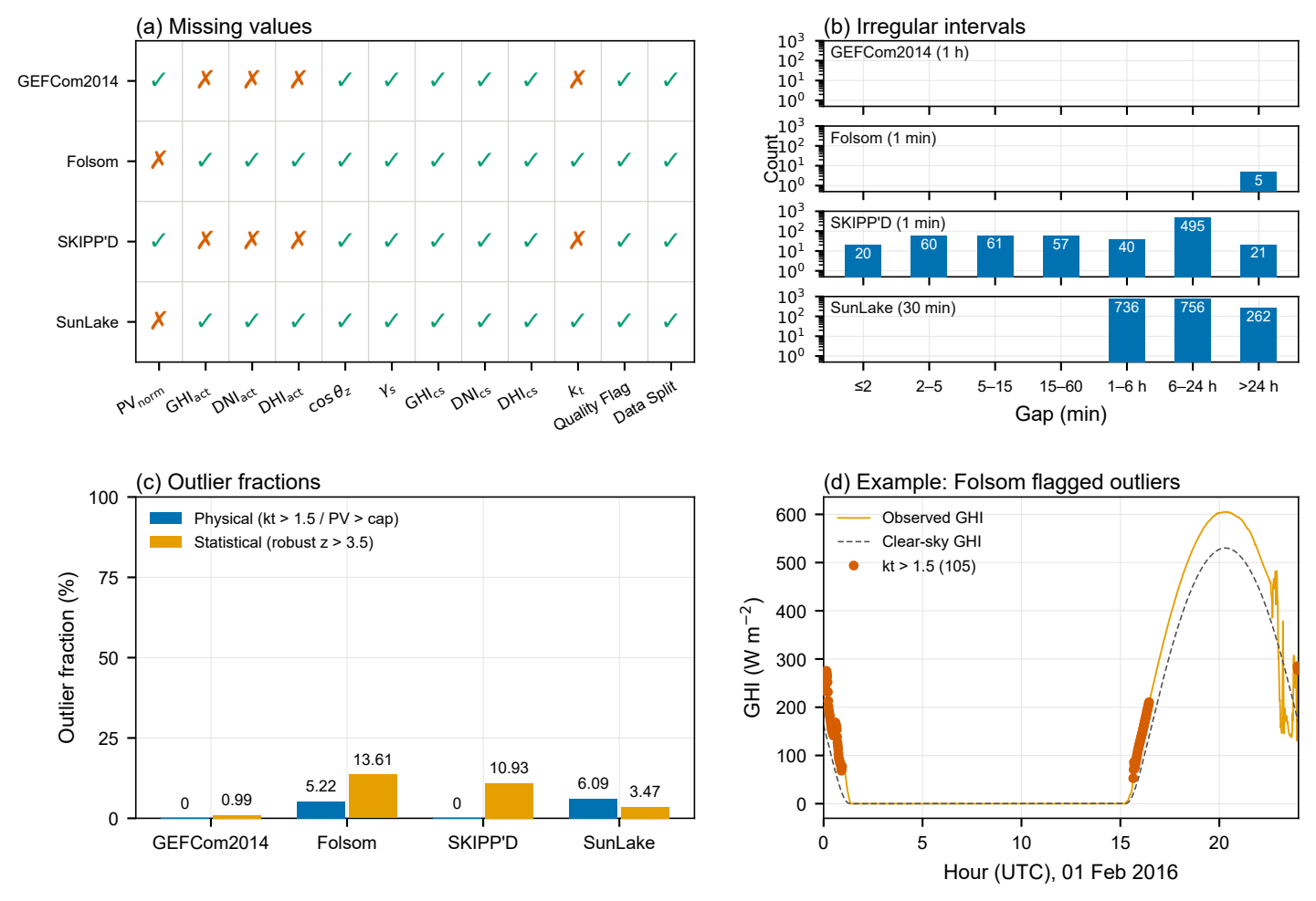

(a) Missing values
GEFCom2014
Folsom
SKIPP'D
SunLake
PV_norm
GHI_act
DNI_act
DHI_act
cos θ_z
γ_s
GHI_cs
DNI_cs
DHI_cs
k_t
Quality Flag
Data Split
(b) Irregular intervals
GEFCom2014 (1 h)
Folsom (1 min)
SKIPP'D (1 min)
SunLake (30 min)
Count
5
20
60
61
57
40
495
21
736
756
262
≤2
2–5
5–15
15–60
1–6 h
6–24 h
>24 h
Gap (min)
(c) Outlier fractions
Physical (kt > 1.5 / PV > cap)
Statistical (robust z > 3.5)
Outlier fraction (%)
0
0.99
5.22
13.61
0
10.93
6.09
3.47
GEFCom2014
Folsom
SKIPP'D
SunLake
(d) Example: Folsom flagged outliers
Observed GHI
Clear-sky GHI
kt > 1.5 (105)
GHI (W m⁻²)
Hour (UTC), 01 Feb 2016